

```
In [1]: import pandas as pd
import sqlite3
```

```
In [2]: orders = pd.read_csv("/home/hanysaa/store_
customers = pd.read_csv("/home/hanysaa/sto
```

```
In [3]: orders.head()
```

```
Out[3]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle;} .dataframe tbody tr th {
vertical-align: top;} .dataframe thead th { text-align:
right;} </style> <thead> <th></th> <th>order_id</th>
<th>customer_id</th> <th>product</th>
<th>category</th> <th>amount</th>
<th>order_date</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th>
</tbody> <tbody></tbody>
```

5000	4	paper	office	14.09	2026-05-25
------	---	-------	--------	-------	------------

5001	15	planner	office	51.86	2026-05-08
------	----	---------	--------	-------	------------

5002	34	shelf	furniture	116.25	2026-06-03
------	----	-------	-----------	--------	------------

5003	2	granola	snacks	19.72	2026-05-19
------	---	---------	--------	-------	------------

5004	17	monitor	electronics	284.80	2026-05-10
------	----	---------	-------------	--------	------------

```
In [4]: customers.head()
```

```
Out[4]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th>
<th>customer_id</th> <th>name</th> <th>city</th>
<th>signup_year</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th>
</tbody> <tbody></tbody>
```

1	Ana Lopez	Venice	2021
2	Ben Kim	Culver City	2023
3	Chloe Nguyen	Santa Monica	2021
4	Dev Patel	Los Angeles	2026
5	Elena Garcia	Culver City	2022

```
In [5]: orders.head()
```

```
Out[5]: <style scoped> .dataframe tbody tr th:only-of-type {
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5004	17	monitor	electronics	284.80	2026-05-10

```
In [6]: con = sqlite3.connect("store.db")
```

```
In [7]: orders.to_sql("orders", con, index=False,
```

```
Out[7]: 200
```

```
In [8]: customers.to_sql("customers", con, index=F
```

```
Out[8]: 40
```

```
In [9]: pd.read_sql("SELECT name FROM sqlite_maste
```

```
Out[9]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>name</th>
</thead> <tbody> <th>0</th> <th>1</th> </tbody>
<tbody></tbody>
```

orders

customers

```
In [10]: pd.read_sql("SELECT * FROM orders LIMIT 5")
```

```
Out[10]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>order_id</th>
<th>customer_id</th> <th>product</th>
<th>category</th> <th>amount</th>
<th>order_date</th> </thead> <tbody> <th>0</th>
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5004	17	monitor	electronics	284.80	2026-05-10

```
In [11]: pd.read_sql("""  
SELECT *  
FROM orders  
WHERE amount > 100  
ORDER BY amount DESC  
""", con)
```

```
Out[11]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
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<th>customer_id</th> <th>product</th>
<th>category</th> <th>amount</th>
<th>order_date</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th>
<th>...</th> <th>79</th> <th>80</th> <th>81</th>
<th>82</th> <th>83</th> </tbody> <tbody>
</tbody>
```

5153	20	dock	electronics	499.00	2026-05-25
5140	30	headset	electronics	295.36	2026-05-26
5047	901	headset	electronics	293.79	2026-05-15
5150	903	monitor	electronics	293.35	2026-06-06
5014	6	monitor	electronics	287.83	2026-05-22
...	...	...	...	...	...
5107	38	shelf	furniture	110.31	2026-05-17
5046	13	shelf	furniture	108.47	2026-05-28
5009	14	chair	furniture	107.20	2026-05-24
5007	36	dock	electronics	103.77	2026-05-21

5023	9	desk	furniture	100.27	2026-06-05
------	---	------	-----------	--------	------------

<p>84 rows × 6 columns</p>

```
In [12]: pd.read_sql("""
SELECT category,
        COUNT(*) AS n,
        AVG(amount) AS avg_amount
FROM orders
GROUP BY category
ORDER BY avg_amount DESC
""", con)
```

```
Out[12]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle;} .dataframe tbody tr th {
vertical-align: top;} .dataframe thead th { text-align:
right;} </style> <thead> <th></th>
<th>category</th> <th>n</th>
<th>avg_amount</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> </tbody> <tbody>
</tbody>
```

electronics	60	190.569500
-------------	----	------------

furniture	46	146.251522
-----------	----	------------

office	51	30.132549
--------	----	-----------

snacks	43	14.619767
--------	----	-----------

<p>the total orders is correct which is 200 orders,however, electronics earn the highest avergae amount which is 190.56 wirh a total orders equal 60</p>

```
In [13]: pd.read_sql("""
SELECT c.city,
        SUM(o.amount) AS total_spend
FROM orders o
JOIN customers c
        ON o.customer_id = c.customer_id
GROUP BY c.city
ORDER BY total_spend DESC
""", con)
```

```
Out[13]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>city</th>
<th>total_spend</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th>
<th>5</th> </tbody> <tbody></tbody>
```

Los Angeles	5862.99
-------------	---------

Culver City	5316.88
-------------	---------

Santa Monica	4017.93
--------------	---------

Inglewood	2718.25
-----------	---------

Torrance	1037.25
----------	---------

Venice	462.36
--------	--------

```
In [14]: pd.read_sql("""
SELECT COUNT(*) AS order_count
FROM orders
""", con)
```

```
Out[14]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th>
<th>order_count</th> </thead> <tbody> <th>0</th>
</tbody> <tbody></tbody>
```

200

```
In [15]: pd.read_sql("""
SELECT COUNT(*) AS joined_count
FROM orders o
JOIN customers c
      ON o.customer_id = c.customer_id
""", con)
```

```
Out[15]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th>
<th>joined_count</th> </thead> <tbody> <th>0</th>
</tbody> <tbody></tbody>
```

195

<p>since we use inner join ,we only shwon a 195 rows which is less 5 rows to the order table ,because the custmoer_Id in the oder table don't match the customer_id on Joined table.</p>

```
In [16]: city_spend = pd.read_sql("""
SELECT c.city,
      SUM(o.amount) AS total_spend
FROM orders o
JOIN customers c
      ON o.customer_id = c.customer_id
```

```
GROUP BY c.city
ORDER BY total_spend DESC
""", con)

city_spend
```

```
Out[16]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
vertical-align: top; } .dataframe thead th { text-align:
right; } </style> <thead> <th></th> <th>city</th>
<th>total_spend</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th>
<th>5</th> </tbody> <tbody></tbody>
```

Los Angeles	5862.99
-------------	---------

Culver City	5316.88
-------------	---------

Santa Monica	4017.93
--------------	---------

Inglewood	2718.25
-----------	---------

Torrance	1037.25
----------	---------

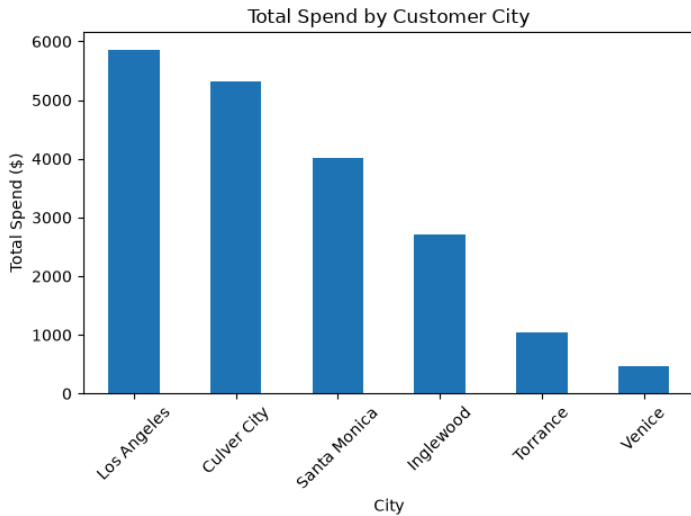
Venice	462.36
--------	--------

```
In [17]: import matplotlib.pyplot as plt
```

```
city_spend.plot(
    kind="bar",
    x="city",
    y="total_spend",
    legend=False
)

plt.title("Total Spend by Customer City")
plt.xlabel("City")
plt.ylabel("Total Spend ($)")
plt.xticks(rotation=45)
```

```
plt.tight_layout()  
plt.show()
```



```
In [18]: pd.read_sql("""  
SELECT product,  
          AVG(amount) AS avg_amount  
FROM orders  
GROUP BY product  
ORDER BY avg_amount DESC  
""", con)
```

```
Out[18]: <style scoped> .dataframe tbody tr th:only-of-type {
vertical-align: middle; } .dataframe tbody tr th {
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<th>avg_amount</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th>
<th>5</th> <th>6</th> <th>7</th> <th>8</th>
<th>9</th> <th>10</th> <th>11</th> </tbody>
<tbody></tbody>
```

headset	213.143478
---------	------------

monitor	186.748824
---------	------------

dock	167.857000
------	------------

shelf	159.497368
-------	------------

chair	152.348824
-------	------------

desk	110.719000
------	------------

planner	31.725385
---------	-----------

pens	29.889000
------	-----------

paper	29.252778
-------	-----------

tea	15.500714
-----	-----------

granola	14.364444
---------	-----------

coffee	13.916364
--------	-----------

<p>Which product has the highest average order amount?</p>

<p>headset the highest average order amount which is 213.14 per oderes.</p>

<p>Ethics note by providing my nootbook publicly,I will be sure to note shae the shore customer_id Data base,which contains the full name of the customers and there full address or anyother personal information.I will only share the publick information about spending moeny for which cataegory or which city has the most orders.</p>

In []: